7.BINARY SEARCH:

```c
#include <stdio.h>

int main() {
    int a[100], n, key, low, high, mid;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter sorted elements:\n");
    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    printf("Enter element to search: ");
    scanf("%d", &key);

    low = 0;
    high = n - 1;

    while (low <= high) {
        mid = (low + high) / 2;

        if (a[mid] == key) {
            printf("Element found at position %d", mid + 1);
            return 0;
        }
        else if (key < a[mid])
            high = mid - 1;
        else
```

```
            low = mid + 1;
        }
    }

    printf("Element not found");

    return 0;
}
```

OUTPUT: